

u4-lasers

n01-characteristics_of_lasers

Characteristics

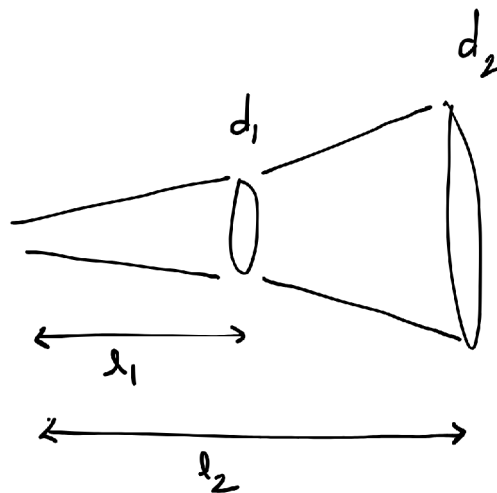
1. Monochromaticity : It is highly monochromatic than the ordinary light.

$$\Delta\lambda = \frac{-c}{\nu^2} \Delta\nu$$

2. Divergence : It has low divergence.

- angle of divergence

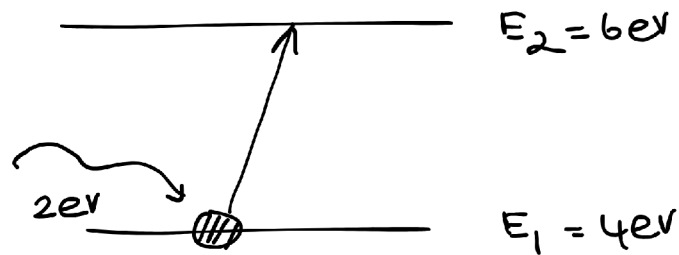
$$(\phi) = \frac{\text{radius}}{\text{distance covered}}$$



$$\phi = \frac{d_2 - d_1}{2(l_2 - l_1)}$$

3. Coherence : It has high coherence compared than ordinary light.
4. Intensity (or) Brightness : It has high brightness compared to ordinary light.

Absorption: Absorbing energy and entering higher energy levels.

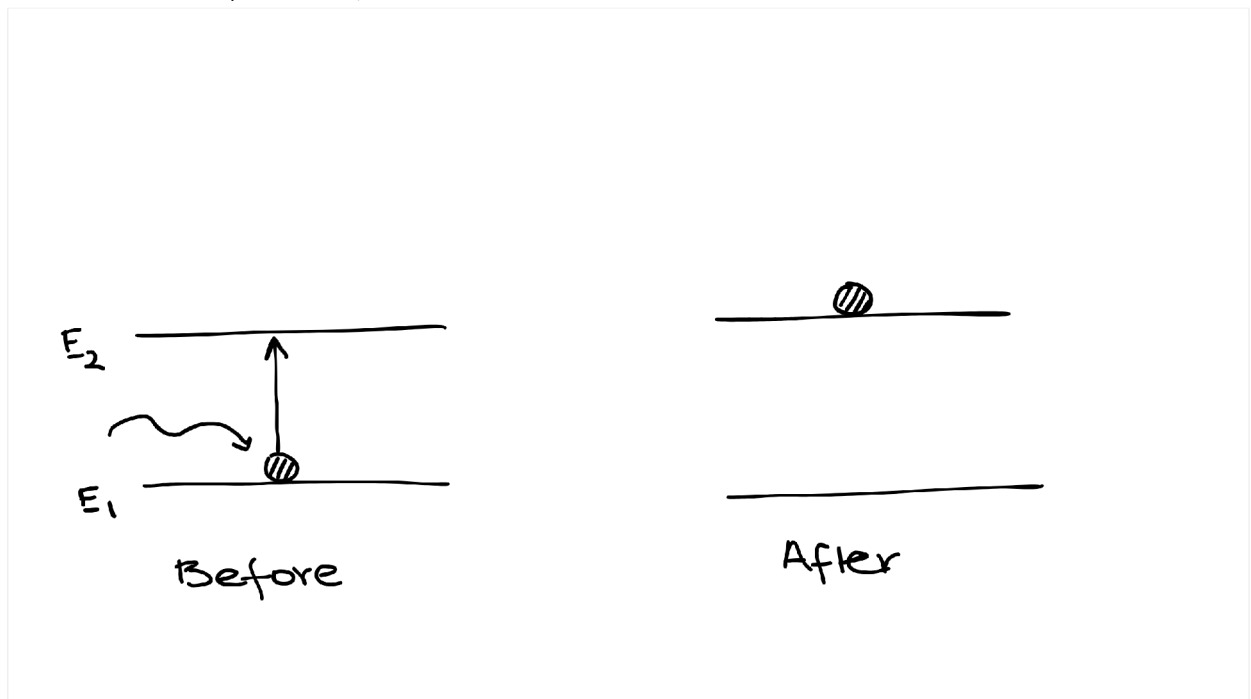


nO2-interaction_of_light_with_matter

In general radiation interaction with matter in 3 ways.

1. Stimulated absorption of radiation:

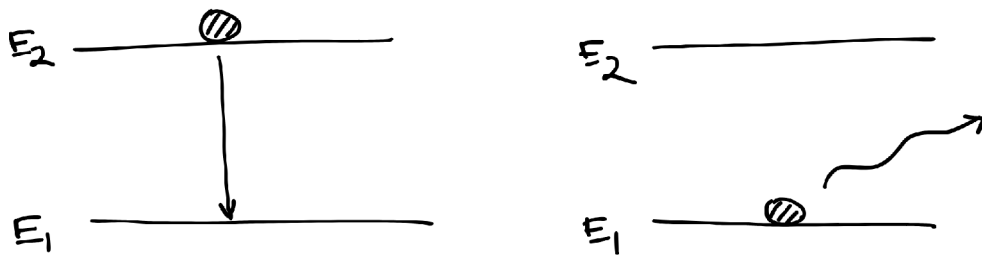
Let E_1 and E_2 are the two energy levels in which E_1 is lower state and E_2 is excited state. Let N_1 and N_2 are the number of atoms per unit volume in E_1 and E_2 respectively.



The incident radiation having energy equal to the difference of E_1 and E_2 . Now photon in incident radiation triggers atom in the lower energy level to absorb the energy and transits to the excited state E_2 as shown in the figure.

2. Spontaneous emission of radiation:

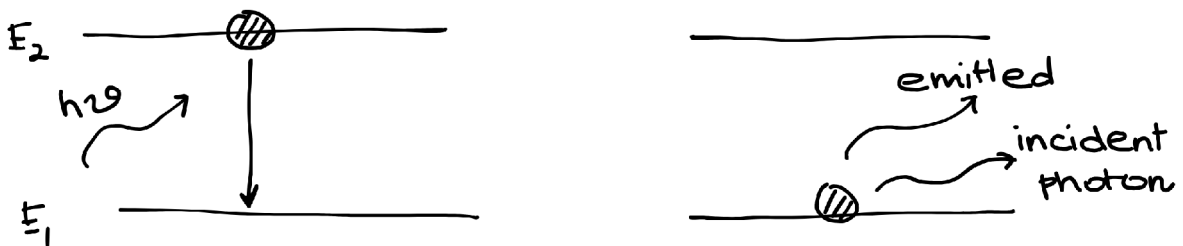
When the atom is excited to higher energy state, it can't stay for a long time there. The life time of excited state is approximately $10^{-8}s$. After this time, the atom de-excites to E_1 by releasing a photon as shown in figure.



This process is called spontaneous emission of radiation and it takes place in ordinary light sources.

3. Stimulated emission of radiation:

When the excited atom comes back to the ground state, by getting triggered by incident photon and emits 2 photons as shown in the figure. This process is called stimulated emission of radiation.



This process is more dominated in lasers and emitted photons are in the same direction having same frequency.

n03-einstein_coefficients

Consider E_1 and E_2 are two energy levels whose population densities are N_1 and N_2 respectively. Let $\rho(\nu) \rightarrow$ incident radiation density (no of photons per unit volume).

The incident light excites ground state atoms to energy level E_2 by giving its energy thus the no of stimulated absorption depend on $\rho(\nu)$ and N_1 .

no of stimulated absorption = $B_{12}N_1\rho(\nu)$

where B_{12} is the einstein's coefficient for stimulated absorption of light.

When the atom is in an excited state it comes to the ground state by spontaneous or stimulated emissions.

no of spontaneous emission $\propto N_2$

$$= A_{21}N_2 \dots (ii)$$

where A_{21} is the einstein's coefficient for spontaneous emissions.

$$\frac{1}{A_{21}} \text{ gives the lifetime of the excited state.}$$

no. of stimulated emission $\propto N_2$

$$\begin{aligned} &\propto \rho(\nu) \\ &= B_{21}N_2\rho(\nu) \dots (iii) \end{aligned}$$

where B_{21} is the einstein's coefficient for stimulated emission.

To obtain the relation between einstein's 3 coefficients, let us assume that the upward and downward transition is equal at thermal equilibrium.

upward transition = downward transition

$$B_{12}N_1\rho(\nu) = A_{21}N_2 + B_{21}\rho(\nu)N_2$$

$$\rho(\nu)(B_{12}N_1 - B_{21}N_2) = A_{21}N_2$$

$$\rho(\nu) = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2}$$

Divide numerator and denominator by $B_{21}N_2$

$$\rho(\nu) = \frac{\frac{A_{21}N_2}{B_{21}N_2}}{\frac{B_{12}N_1}{B_{21}N_2} - 1}$$

$$\rho(\nu) = \frac{\frac{A_{21}}{B_{21}}}{\frac{B_{12}N_1}{B_{21}N_2} - 1} \dots (iv)$$

According to the boltzmann distribution law

$$N_1 = N_0 e^{-E_1/kT} \dots (v)$$

$$N_2 = N_0 e^{-E_2/kT} \dots (vi)$$

$$\frac{N_1}{N_2} = \frac{N_0 e^{-E_1/kT}}{N_0 e^{-E_2/kT}}$$

$$\frac{N_1}{N_2} = e^{(E_2 - E_1)/kT}$$

$$\frac{N_1}{N_2} = e^{h\nu/kT} \dots (vii)$$

from eqn (iv) and (vii)

$$\rho(\nu) = \frac{A_{21}/B_{21}}{\frac{B_{12}}{B_{21}} e^{h\nu/kT} - 1} \dots (viii)$$

Generally the emitted radiation is in the form of electromagnetic radiation that resembles with blackbody radiation. Therefore the radiation density should be Plancks distribution.

$$\rho(\nu) = \frac{8\pi h\nu^3}{c^3} \left[\frac{1}{e^{h\nu/kT} - 1} \right] \dots (ix)$$

Comparing (viii) and (ix)

$$\frac{B_{12}}{B_{21}} = 1 \implies B_{12} = B_{21} \dots (x)$$

This implies that the upward and downward transitions are equal.

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$$

The above equation indicates that the spontaneous emission is more if the frequency is high. This is why it is very difficult to construct x-ray lasers.

n04-light_amplification

Light can be amplified only if stimulated emissions are dominant.

i) Stimulated emission > Spontaneous emission

$$\frac{B_{21}N_2\rho(\nu)}{A_{21}N_2} = \frac{B_{21}}{A_{21}}\rho(\nu)$$

$$\dots \text{ (referencing equation } \frac{A_{21}}{B_{21}}) = \frac{c^3}{8\pi h\nu^3} \rho(\nu)$$

If $\rho(\nu)$ is high there is a chance of getting stimulated absorption also. This will not guarantee high stimulated emission.

If life time of excited state is high then stimulated emissions dominate over spontaneous emissions.

ii) Stimulated emission > Stimulated absorption

$$\frac{B_{21}N_2\rho(\nu)}{B_{12}N_1\rho(\nu)} = \frac{N_2}{N_1}$$

since $B_{12} = B_{21}$

$$N_2 > N_1$$

This state is called Population Inversion.

n05-difference_between_spontaneous emissions_and_stimulated_emissions

1. Spontaneous emissions:

- this takes place when the atom de-excites to lower state by itself.
- This is a slow process when compared to stimulated emission.
- Independent of $\rho(\nu)$.
- photons are emitted in any direction.
- Emitted photons are non-monochromatic and non-coherent.
- this process is observed in ordinary light.

2. Stimulated emissions:

- This takes place when the excited atom is triggered by the incident radiation.
- this is a fast process when compared to spontaneous emission.
- Dependent of $\rho(\nu)$.
- photons are emitted in a single direction.
- emitted photons are highly monochromatic and highly coherent.

- This process is observed in laser light.

— 000 — E_2

— 0000000 — E_1

$N_1 > N_2$

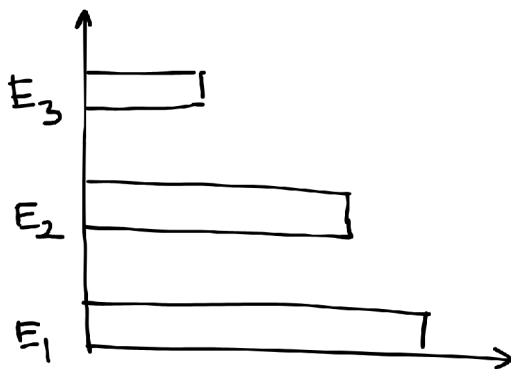
thermal
equilibrium

— 00000000 — E_2

— 000 — E_1

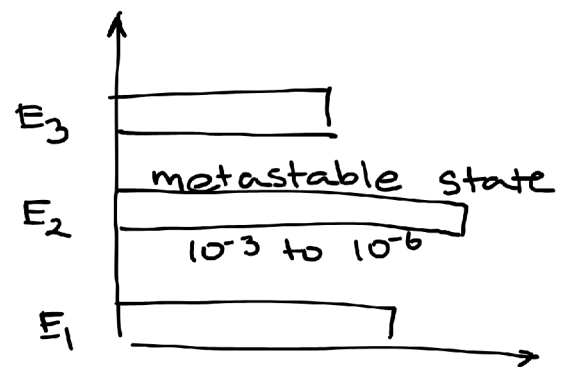
$N_2 > N_1$

non-equilibrium



$N_2 < N_1$

distribution in
thermal equilibrium



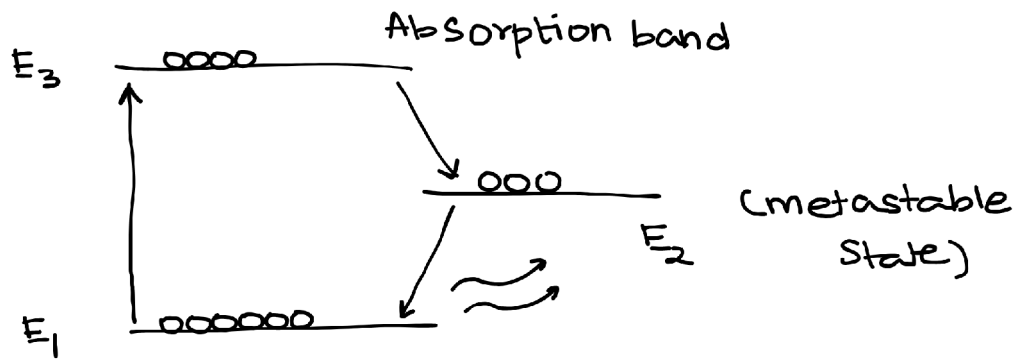
distribution in
non-equilibrium

n06

Metastable state

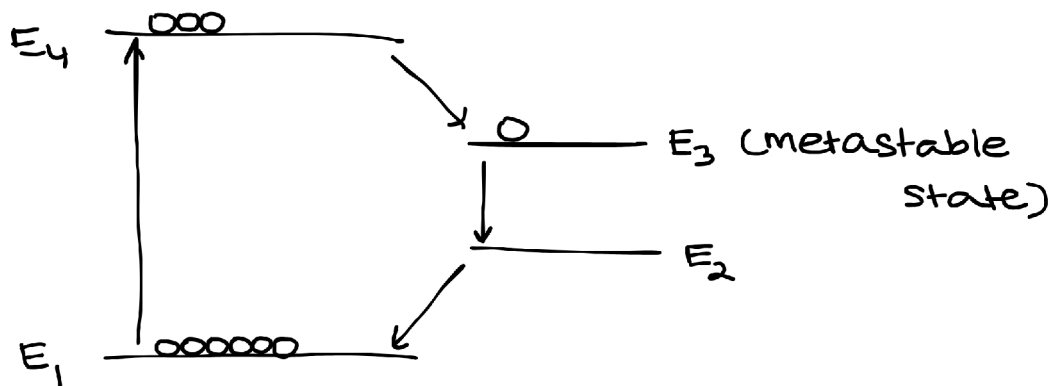
The metastable state is an intermediate state which have longer lifetime (10^{-3} to 10^{-6} s) when compared to the excited state.

- If the metastable state is not there, then laser action is not at all possible, since there is no chance of population inversion.



Disadvantages

More energy required to pump E_1 to E_3 , since E_2 to E_1 results in greater re-absorption losses due to high E_1 population



n07-pumping

Pumping is the process of sending atoms to higher energy states to achieve population inversion.

i) Optical pumping

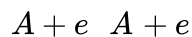
- In optical pumping, the light source is used to supply energy for the excitation of atom to higher states.
- This is observed in ruby laser.

ii) Electric discharge method

- In this method, electrons are produced in an electrical discharge tube and these electrons are accelerated by supply of strong electric field.
- The fast moving electron then collides with the atom in the ground state and excites the atom to higher energy state by giving its energy.
- this type of pumping is observed in gaseous lasers (Argon-ion laser).

iii) Inelastic atom-atom collision

- In this method, a mixture of two gases A and B are considered.
- the gases are chosen in such a way that the excited states of two gases should be identical.
- In the first step, the ground state atom of gas 'A' bombards with the fast moving electrons and excited to higher state.



- In the second step excited 'A' atoms collides inelastically with the gas 'B' atoms.
- then the atoms of gas 'B' gains energy and move to excited state.



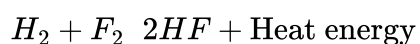
- this type of pumping is seen in Helium-neon laser.

iv) Direct conversion

- In this method direct conversion of electric energy into optical energy takes place.
- In a semiconductor laser like GaAs, the re combination of the electron and holes takes place, when p-n junction is forward biased. This recombination produces laser light.

v) Chemical Reaction

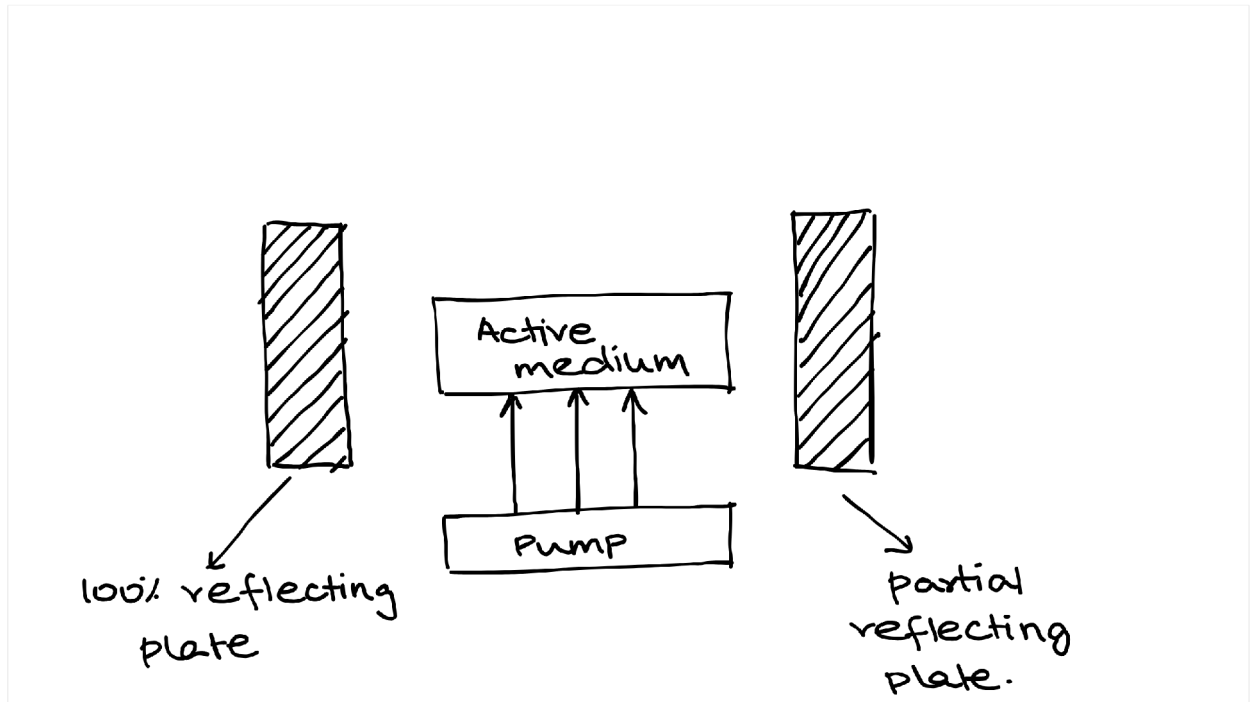
- In chemical lasers there is no need of other excited mechanism.
- The heat produced during the chemical reaction solves the purpose.



n08-components_of_lasers

1. Active medium
2. Pump

3. Optical cavity/resonate



Active Medium - Active medium is the medium where stimulated emission takes place.

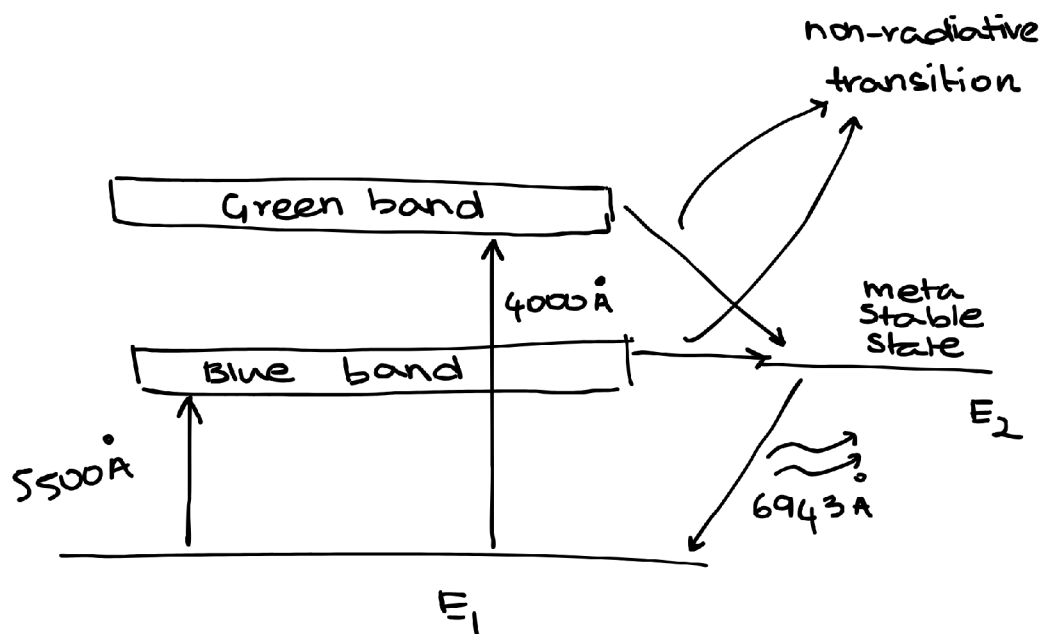
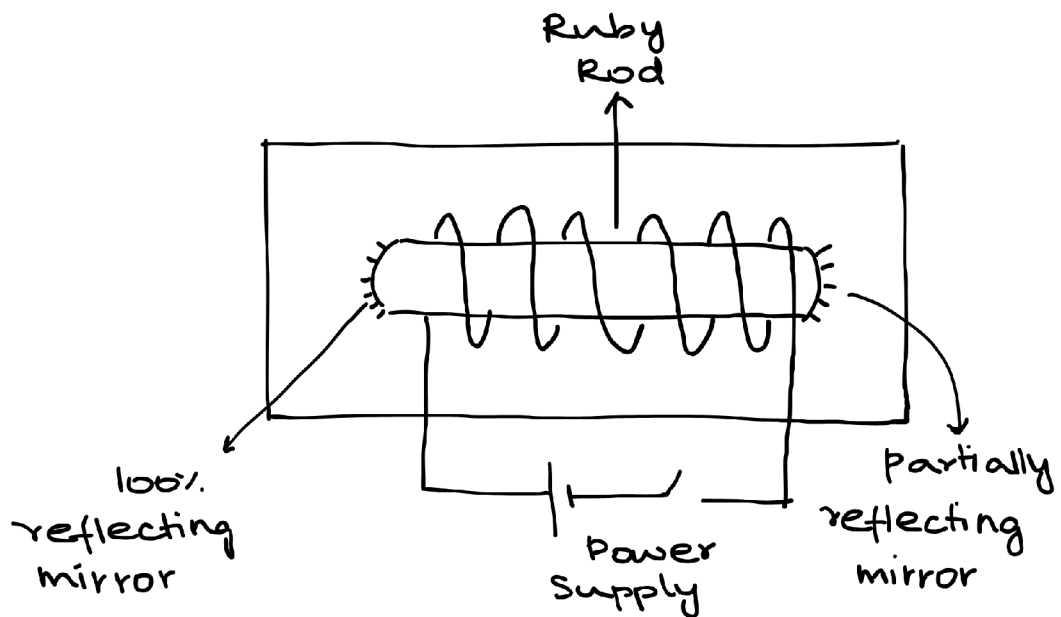
Pump - The source of energy which supplies energy to active medium to produce a population inversion

n09-types_of_lasers

The lasers are of mainly three types based on the material used.

1. Solid State Laser
Ex: Ruby, Nd:YAG laser
2. Gaseous State Laser
Ex: He-Ne, CO_2
3. Semiconductor Laser
Ex: GaAs laser and InP laser.

n10-ruby_laser



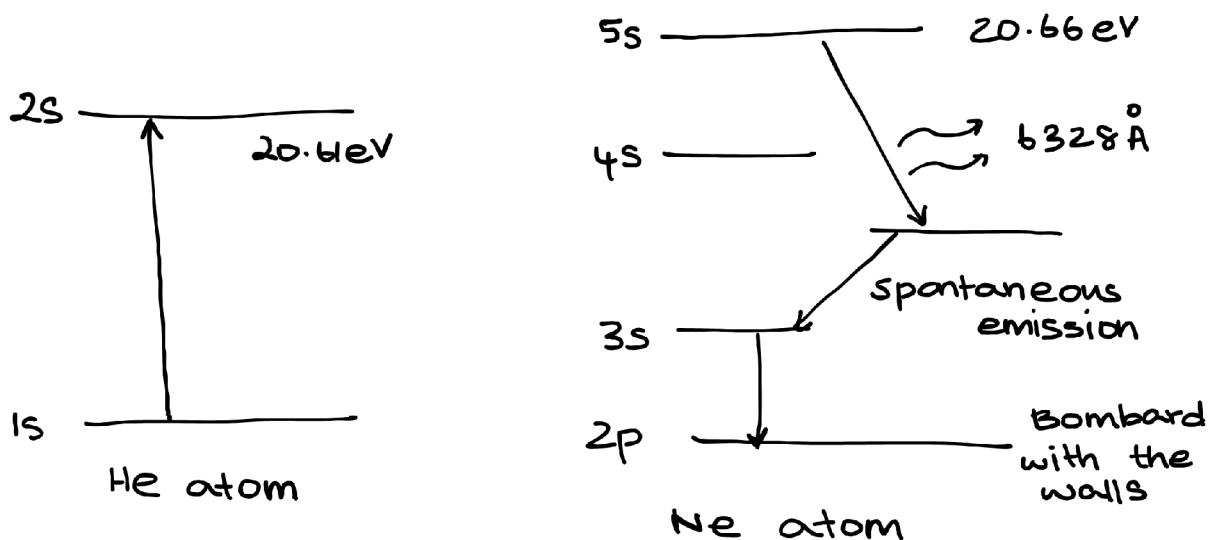
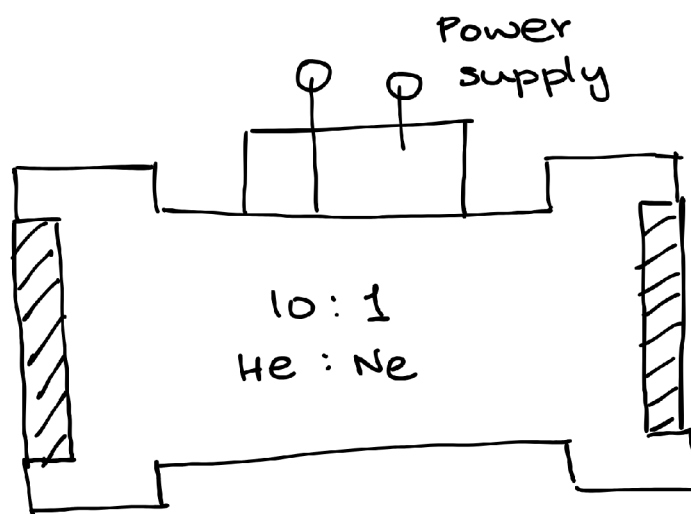
1. more pumping required
 2. only green and blue is used, remaining is wasted.
- once the power supply is on, the xenon flash lamp sends one pulse light which supplies energy to the Cr^{3+} ions.
 - chromium ions have the absorption band of green and blue, therefore these two components of flashlight are used to excite Cr^{3+} ion to higher excited state.
 - the excited Cr^{3+} ion reaches the metastable state through non radiation transition.
 - Since the lifetime of E_2 is high, population inversion occurs between E_2 and E_1 .

- A laser of 6943Å is emitted from partially reflecting mirror.
- Since the lower laser level is the ground state, more pump power is required to get laser light.

Feature

- It is a three level laser
- Cr^{3+} ions are active senders
- High pump power is required
- And output the pulsed mode not in continuous mode

n11-helium-neon_laser



Energy Level Diagram:

- He atom: $1s \rightarrow 2s$ (20.61eV)
- Ne atom: 20.66eV
- Spontaneous emission (6328Å)
- $3s \rightarrow 2p$
- $2p \rightarrow 1s$ (Bombard with the walls)

n12-semiconductor_laser

It is a specially fabricated p-n junction which emits coherent light when it is forward biased.

Principle

- p-n junction is formed when we combine p-type material with n-type material.
- the width of depletion region reduces when it is forward biased.
- the electron from conduction band crosses the barrier and recombines with hole in p-type.
- this recombination emits laser light across junction.
- there are two types of semiconductor laser.

i) Homojunction semiconductor laser

- Is a type of laser which uses the same type of semiconductor material on both sides of p-n junction.
- Ex: GaAs laser.

ii) Heterojunction semiconductor laser

- It is a laser which makes use two different semiconductor materials on both sides of pn junction.
- Ex: A semiconductor which is having GaAs and GaAlAs on other side.